

Leon Lin

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Education

Georgia Institute of Technology

Atlanta, GA

Major: Computer Science, Master's of Science

August 2026 - May 2028

University of California, San Diego

La Jolla, CA

Major: Mathematics-Computer Science, Bachelor's of Science

September 2023 - June 2026

Laney College

Oakland, CA

Major: Computer Science

August 2021 - May 2023

Experience

Windows OS Benchmark - Graduate OS research project | Report

Oct. 2025 - Dec. 2025

- Built a C++17 microbenchmark suite for Windows using **RDTSC**, hardware fences, warm-up runs, and repeated trials to measure CPU, memory, filesystem and TCP operations at cycle-level precision
- Designed repeatable experiments to analyze system-call latency, context switches, cache and DRAM behavior, page faults, and filesystem performance while controlling measurement overhead and environmental noise
- Benchmarked Windows loopback and **AWS EC2** networking, measuring RTT, throughput, and connection setup/teardown latency across multiple workloads and message sizes

Project Experience

Autopilot

Feb. 2026 - March 2026

- Led a **team of 3** to build a **Yelp-like microservice system** in Go by implementing service endpoints and **gRPC-based communication**, and conducted latency benchmarking with **load generator(wrk2)** to simulate production workloads and identify system bottlenecks on GCP's Virtual Machines
- Reduced **P99 tail latency by 70%** for service requests by implementing an **Autopilot scaling service** that dynamically reallocates CPU cores across microservices based on **real-time queue depth**

Distributed GPT-2 System

June 2026 - July 2026

- Trained a **124M-parameter GPT-2 language model** on **~10B FineWeb-Edu tokens** with a **524K-token global batch size** by building a PyTorch training pipeline with sharded dataloading and cosine learning-rate scheduling
- Increased single-GPU training throughput **10.6× from ~16K to ~170K tokens/sec** and reduced peak GPU memory usage **2.3× from ~35GiB to ~15GiB** by applying TF32 precision, BF16 autocasting, torch.compile, PyTorch FlashAttention, vocabulary-size alignment and iterative performance benchmarking
- Scaled training throughput to **~1.4M tokens/sec on 8×A100-40GB GPUs** by implementing Distributed Data Parallel training with torchrun, gradient accumulation, rank-aware shard loading and Modal cloud infrastructure

TritonTube

May 2025 - June 2025

- Reduced video upload latency by **97%** for 475 MPEG-DASH segments using a **thread pool** to parallelize upload operations and improved migration throughput by **98%** through gRPC batch transfers to minimize network I/O
- Implemented a distributed video streaming platform using consistent hashing for **horizontal scaling** that supports the dynamic addition and removal of storage nodes and automatically balances client requests across nodes
- Designed a distributed, **fault-tolerant** metadata subsystem using a **multi-node etcd cluster** to achieve **strong consistency** via **Raft-based replication** and maintaining **high availability** under node failures

Skills

Languages/Frameworks: Go, Java, Python, C++, C, CUDA, gRPC, PyTorch

Tools: AWS, GCP, Git, Kubernetes, Docker, Linux, CI/CD

Courses: Graduate Operating Systems, Computer Networks, Distributed Systems, Parallel Computing, Machine Learning Systems, Data Center Systems, Data Structures, Algorithms, Deep Learning